

that support the promiscuous mode to view all the traffic on that interface. This includes the interface configured address and broadcast/multicast traffic. When capturing with a packet analyser in promiscuous mode, on a port of a network switch, some of the traffic through the switch is sent to the port where the capture is done. If a remote machine captures packets and sends them to a machine running Wireshark using the TZSP protocol, Wireshark dissects those packets, so that it can analyse them on the remote machine.

Features:

- Live packet capture from a network interface
- Opens files containing packet data with *tcpdump/WinDump* and a number of packet capture programs
- Imports packets from text files containing hex dumps of packet data
- Displays packets with detailed protocol information and exports all packets in a number of capture file formats
- Has coloured packet displays based on filters and creates various statistics

Official website: www.wireshark.org

Latest version: 2.2.0

AMANDA Network Backup

AMANDA (Advanced Maryland Automatic Network Disk Archiver) is an open source computer archiving tool that backs up data residing on multiple computers onto one particular system. It makes use of a client-server model, where the server makes a direct link with every client on the network to perform data backups at a scheduled time. AMANDA Network Backup was developed at the University of Maryland, and is available as an open source-cum-free edition and also as an enterprise edition.

AMANDA runs on over a million servers and desktops running various versions of Linux, UNIX, BSD and Microsoft Windows.

Features:

- AMANDA makes the backing up of networks simple, as a single server is set up to back up multiple clients on a tape or a disk based storage system.
- It has the unique capability of writing backups to tape and disk simultaneously. The same data could be available online for quick restore from disk and even for disaster recovery.
- It uses native dump and GNU tar utilities. Data can be recovered with native utilities, regardless of whether AMANDA is installed or not.
- AMANDA provides proper security. Encryption on the client ensures the security of data in transmission and data is stored on the server in an encrypted form. It supports up to 4096-bit keys with public key cryptography as well as 256-bit AES encryption.
- It has a unique scheduler to optimise backup levels for

different clients in such a way that the total backup time is about the same for every backup run. It frees systems administrators of the task of guessing the rate of data change in their environments.

- AMANDA is highly stable and robust, and has a very large and growing community.

Official website: www.amanda.org

Latest version: 3.3.9

Oracle VirtualBox

Oracle VirtualBox is a free, open source and powerful x86 and AMD64/Intel64 virtualisation software for enterprises as well as home use, developed by Oracle Corporation. It can be installed on a number of operating systems like Linux, OS X, Windows and Open Solaris.

It supports the installation of multiple guest OSs under a single host operating system. Every virtual machine (VM) can be started, paused and even stopped independently within its own VM. The end users can configure each VM with either software-based virtualisation or hardware assisted virtualisation.

Software based virtualisation: This mode supports 32-bit guest OSs, which run in Rings 0 and 3 of the Intel Ring Architecture. A guest OS is configured in Ring 0, executes in Ring 1, while guest mode code runs in Ring 3.

Hardware assisted virtualisation: This supports both Intel's VT-x and AMD's AMD-V hardware virtualisation.

The system emulates the hard disk in three formats:

- (a) VDI or VirtualBox Disk Image, (b) VMDK — format used by VMware Workstation, and (c) VHD — Windows Virtual PC format.

Features:

- *Paravirtualisation and improved host support:* It has support for the guest OS (Hyper-V on Windows and KVM on Linux). It also includes support for the NDIS6 driver API.
- *Additional instruction set support:* It includes SSE 4.1, 4.2 and AES-NI as well as POPCNT, RDRAND and RDSEED.
- *Disk image encryption:* It encrypts the virtual disk image using AES 256-bit on the physical drive.
- It has USB 3.0 device support and bi-directional drag-and-drop.
- It also has HiDPI support.

Official website: www.virtualbox.org

Latest version: 5.1.6

ClamWin free antivirus software

ClamWin is free, open source antivirus software for Microsoft Windows, which provides a GUI interface to the Clam antivirus engine. It is mainly written in C++ and Python. It features a high detection rate for viruses and spyware, a scanning scheduler and automatic updates for the virus